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Some hospital patients suffer from diarrhoea caused by infection with the bacterium *Clostridium difficile*. The *C. difficile* bacteria release toxins. These toxins cause the diarrhoea.

The toxins damage the cells lining the ileum, causing them to lose their microvilli. The damage to the cells reduces the absorption of the products of digestion and reduces the absorption of water, resulting in diarrhoea.

Explain why the damage to the cells lining the ileum reduces absorption of the products of digestion **and** why this reduces absorption of water.

[3 marks]

Question 4 continues on the next page

Not all patients in hospital with *C. difficile* develop diarrhoea.

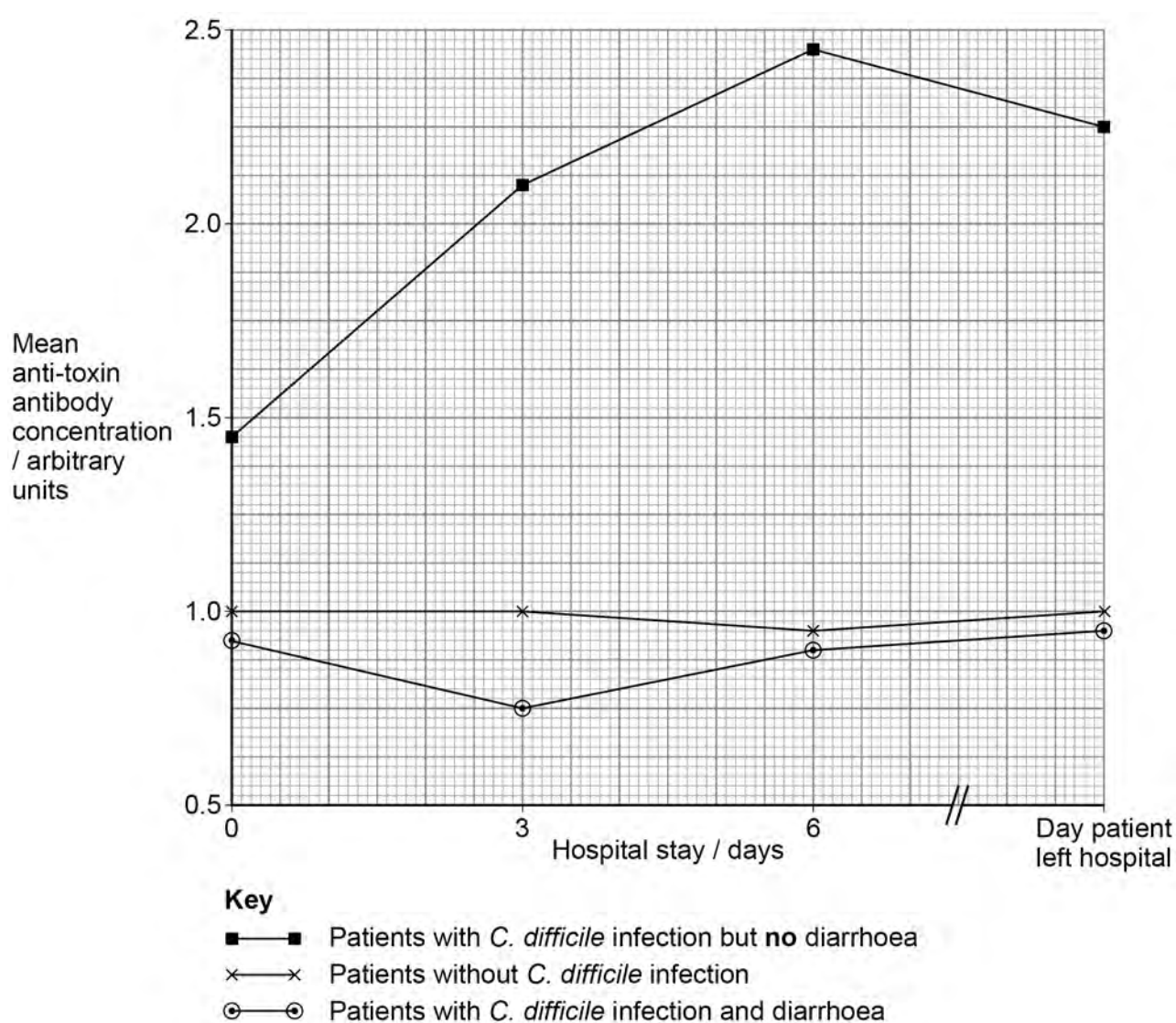
Scientists measured the anti-toxin antibody concentration in hospital patients with and without *C. difficile* infection.

They measured the anti-toxin antibody concentration **four** times:

- on admission to hospital (day 0)
- on day 3
- on day 6
- on the day the patient left the hospital.

Figure 4 shows the scientists' results.

Figure 4



0 4 . 2

The scientists suggest that the anti-toxin antibody could be given to some patients as a form of passive immunity.

Use **Figure 4** to suggest how this passive immunity would work **and** which patients should be offered this anti-toxin antibody.

[3 marks]

0 4 . 3

To be used as passive immunity treatment, the anti-toxin antibody would be injected. If it was given by mouth, it would be digested.

Describe how the anti-toxin antibody would be digested.

[3 marks]

Turn over for the next question

Turn over ►

